computer performance

↳ 3 factors

- instruction count
  number of instructions executed by program
- clock cycle time
  time for one clock period which smallest unit of activity of a processor is executed and runs at constant time
- clock cycles per instruction
  average clock cycles per instruction

Basic set of instructions:

- memory reference — load-store
- logical arithmetic — add, sub, etc.
- conditional branch — beq etc.

STEPS

↳ PC → memory that holds that instruction.

↳ read one/two registers

↳ apply the instruction

Implementation overview

PC → Instruction memory.

sends address of next instr to inst. memory

Instruction memory → registers

sends inst to registers for processing

Registers → Data Memory/

Register → ALU

sends data to ALU.

ALU → Data Memory/

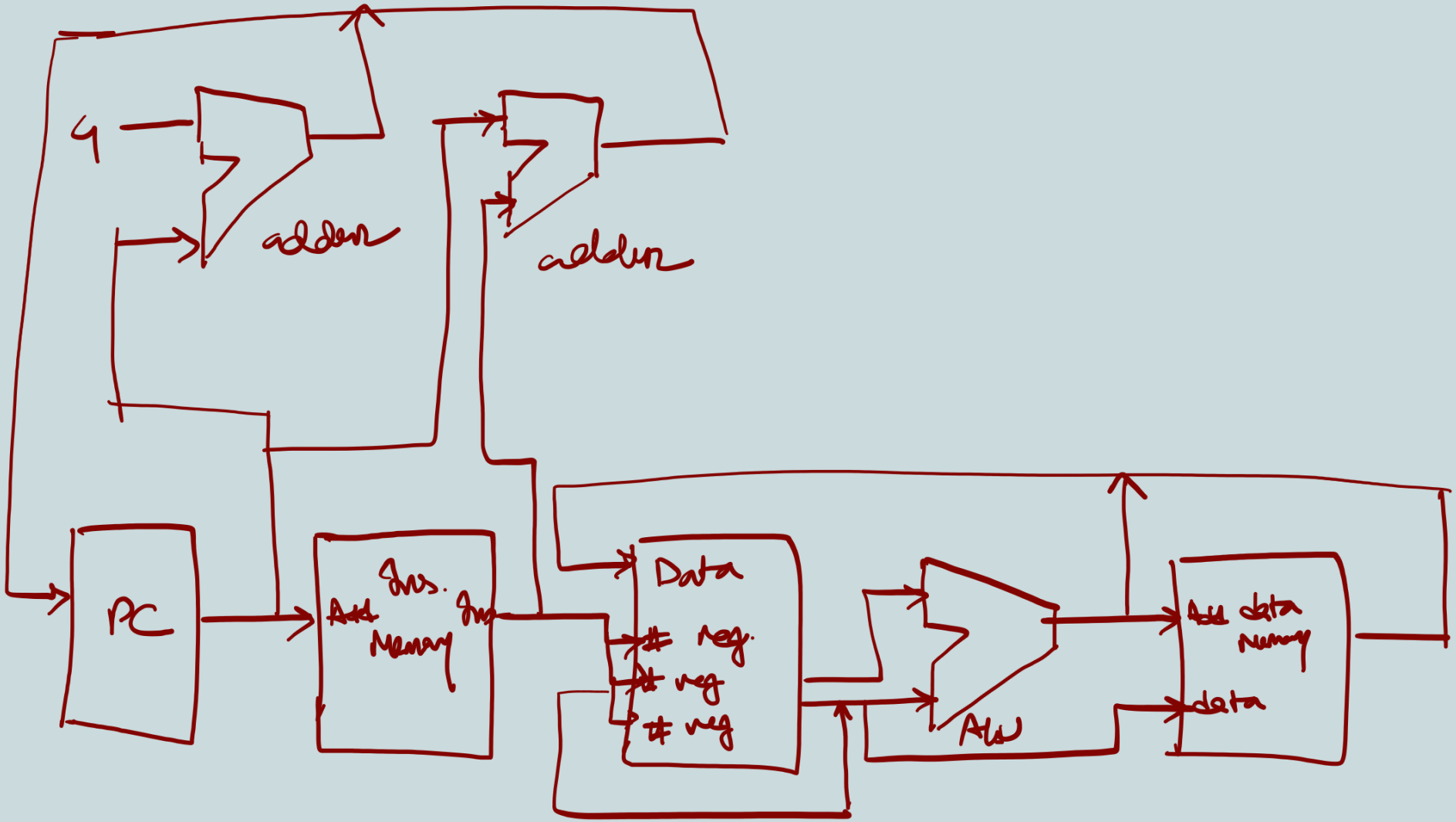
stored for other purposes.

ALU → Registers.

keeps in mind for future purposes

PC → Adder +4 → Next Instruction

→ 2nd Adder → for calculating memory addresses


4
adder
adder
PC
Ins.
Add
Memory
Ins
Data
reg.
reg
reg
ALU
Add data
Memory
data

Multiplexer

control line এর basis এ input select করে

Control unit

instruction input নিয়ে control line set করে

for → functional units
→ multiplexers

① hardware logic (how it operate)
② clocking methodology (how it clocks)

hardware

Combinational elements

- current outputs depend on current inputs.
- same output for same input
- no storage
- AND gate, multiplexer, ALU etc.

Sequential elements

- contains state
- output depend on current & prev inputs
- storage available.
- register/memory etc

clocking methodology

determines the time of when the data is stable and ready to be used based on the timing of the clock in the processor.

clock → data when read/write করা হবে সেটা (যেভাবে বুঝাবে

grab data → read → access data
→ write → change data.

hardware এটা predictable করে
↳ read & writing signals
clock এর উপর depend করে
so mix up হয়না

Clock Edge

transition from low to high

Edge-triggered clocking
↳ clock edge এ এর state
change হয়

sequential logic এ clock edge থাকে
combinational এ set of current input + prev input
= set of current output

signal goes like this

[state 1] → [combinational logic] → [state 2]

no feedback →
এটার state to এরটার state
এ গেলে can't return back to
state 1.

Control signal